

WILL BORDEN

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EDUCATION

- The University of Melbourne · MEng Electrical Engineering** Mar 2026 – Present
Weighted Average Mark: 87.9
- The University of Melbourne · BSci Electrical Engineering Systems** Feb 2022 – Dec 2025
Weighted Average Mark: 87.3 · Dean's Honours List (2022) · Diploma in Languages – French · Academic Exchange, The University of Bordeaux, Aug 2024 – Dec 2024

EXPERIENCE

- Melbourne Space Laboratory** Jul 2025 – Present
Electronics Engineer - Research Assistant Melbourne
- Designed rad-hard thermal control unit for ASA lunar rover, carrying NASA payload to southern lunar pole - deterministic control for 20 heaters, accurate temperature measurement for 60 thermistors.
 - Implemented NVIDIA Jetson-based camera payload with 30 m GSD, launching on Gilmour Space bus in June 2026.
 - Designed and manufactured complex (8+ layer), high-speed, EMI compliant PCBs for use with rad-hard electronic components, including detailed failure mode analysis and error budgets for lunar and orbital environments.
 - Developed Jetson firmware (device-tree config, custom drivers), and RTOS software for MSP430 microcontrollers.
- Esper Satellite Imagery** Feb 2025 – Jul 2025
Electronics Engineer Melbourne
- Designed and validated space-grade OBC and PDN for hyperspectral Earth observation payload launching August 2026, achieving 30 m GSD over 138 spectral bands in both VNIR and SWIR.
 - Managed interface control documents and design reviews with external space providers to ensure compliance with mission and launch provider requirements.
 - Designed and outsourced space-grade wiring harnesses and PCBs; conducted extensive electrical failure mode analysis.
- ARES Rocketry Team** Feb 2022 – Present
Chief Engineer – 2023 - 2024, Lead Systems Engineer – 2025 - Present The University of Melbourne
- Led a multidisciplinary team of 100+ students to design and launch a 30,000 ft sounding rocket, achieving 4th place out of 152 universities at the 2024 Spaceport America Cup.
 - Established ARES' first formal systems engineering framework; authored 50,000+ words of technical documentation and led technical development for ARES' first student-designed hybrid rocket.
 - Created avionics flight computer for parachute deployment and in-flight data logging; selected for podium presentation at the 2023 Spaceport America Cup (20 of 120 submissions). Flown on 10+ missions over 3+ years without failure, including neural-network accelerated MPC air braking system achieving 0.73% apogee error at Mach 1.8.
- QinetiQ Australia** Feb 2023 – Present
Technical Staff – University of Melbourne Creator Space Melbourne
- Supervised and mentored students and professional researchers in a state-of-the-art makerspace, providing expert training in electronics, PCB debugging, reflow soldering, and test equipment operation.
 - Developed and ran bottle rocket outreach program for women and Indigenous students in STEM programs.

TECHNICAL SKILLS

PCB & Circuit Design: 8+ layer PCB design in Altium Designer (7 yrs, started age 16), KiCAD, EagleCAD; analog, high-speed, RF (esp. GNSS systems) & embedded systems; radiation-tolerant design for LEO/lunar; EMI/EMC design practices; SPICE/TINA-TI/QUCS simulation

Embedded Systems: Embedded C/C++ (5 yrs); RTOS frameworks; driver development; PID & Kalman control; FPGA (Verilog/Quartus); protocols: I2C, SPI, UART, USB, CAN, Ethernet, PCIe, MIPI CSI-2; Git, Linux

Systems Engineering: Failure mode & reliability analysis; test campaign design & hardware validation; requirements management; interface control documentation; project leadership (100+ member teams)

Tools & Equipment: Oscilloscopes, function generators, spectrum analyzers, VNAs; Atlassian project management suite; Advanced SMT assembly (BGA, 0201 passives) & hand soldering

AWARDS & HONORS

4th Place (of 152 teams) – Spaceport America Cup 2024 · Podium Presentation Selection – Spaceport America Cup 2023 (20 of 120) · Dean's Honours List, University of Melbourne 2022 · Melbourne Global Scholar's Award 2024 · Alan Gilbert Mobility Scholarship 2024